

```

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```

→ 1
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α 1
β 1
γ 1
δ 1
ε 1
θ 1
λ 1
μ 1
ν 1
π 1
σ 1
η 1
< 1
≥ 1
≠ 1
∈ 1
∞ 1
Σ 1
Δ 1
Ω 1
channel.py
FiberChannel
_physics.py
noise_models.py
L
α
L
Ltotal
TypeVar
Generic
Protocol
Final
channel.py
FiberChannel
AttenuationOnlyFiberChannel
transmittance_array
transmittance_scalar
ChannelSimParams
ContextVar
_physics.py
noise_models.py
-

```

$$T(L) = \frac{P(L)}{P(0)} = 10^{-\alpha L/10},$$

$$(1) \quad \frac{T}{P(L)} \frac{L}{P(0)} \frac{\alpha}{L} L_{total} [dB] = \alpha [dB/km] \times L [km],$$

$$(2) \quad T = \frac{10^{-L_{total}/10}}{1550 \text{ nm}} \alpha \approx 0.2 \text{ dB/km} \cdot 50 \text{ km}$$

$$\eta = \eta_{channel} \cdot \eta_{detector} \cdot \eta_{coupling},$$

$$(3) \quad \eta_{channel} = \frac{T}{\eta_{detector}} \frac{0.1}{0.3} \frac{0.05}{0.15} \frac{\eta_{coupling}}{0.5} \frac{0.3}{0.3}$$